

TrES-3b

R-0290

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180508 · created 2026-07-17T17:34:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458246.843 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.011
Transit depth (Rp/Rs)^2	2.77 +/- 0.38 [%]
Orbital Inclination (inc)	83.03 +/- 0.42
Airmass coefficient 1 (a1)	1.0143 +/- 0.0086
Airmass coefficient 2 (a2)	-0.0116 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	#2 - [311, 282]
Optimal Aperture	3.78
Optimal Annulus	11.77
Transit Duration (day)	0.0645 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.011 vs 0.16309 (deviation 0.26σ)
Duration fit vs published	0.0645 d vs 0.0567 d (deviation 2.45σ)
Mid-time O–C	-2.6 min
Depth SNR	15.1
Residual scatter	0.84 %

Light curve

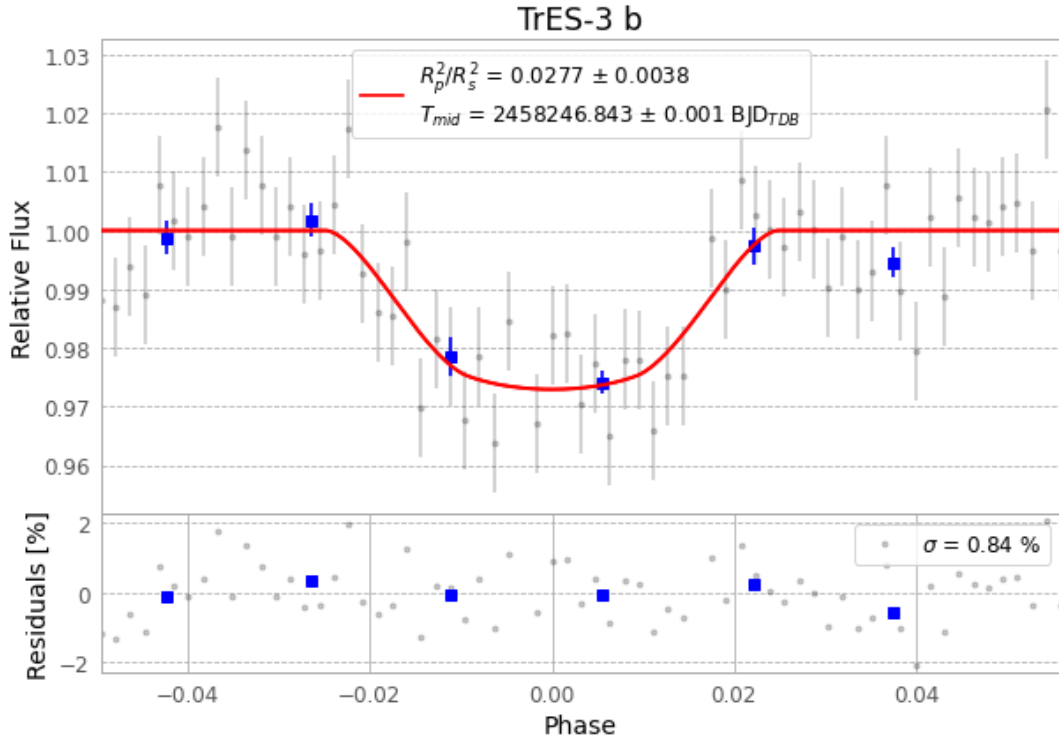


Figure 1 — FinalLightCurve_TrES-3 b_2018-05-07.png (SHA-256 9f7f26c86c4fbd5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 929c7f9562be20f1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

67 FITS frames (44 MB), TRES-3180508063612.FITS → TRES-3180508095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180508.log (001b449745a3...), FinalLightCurve_TrES-3 b_2018-05-07.png (9f7f26c86c4f...), FinalLightCurve_TrES-3 b_2018-05-07.csv (0174ab1c4f11...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 17:34Z.